

National Biodefense Analysis and Countermeasures Center



Science and Technology

UNIQUE AND ESSENTIAL FOR BIODEFENSE

The National Biodefense Analysis and Countermeasures Center (NBACC) is a central element of the United States Biodefense Enterprise, **located at Ft. Detrick, MD.**

NBACC was the first national laboratory created by DHS. It is the nation's only high-containment laboratory that supports homeland security missions while operating at biosafety levels 2, 3, and 4. It is the principal federal facility for laboratory-based biological threat assessments and bioforensics.

NBACC is one of only 12 operational BSL-4 laboratories in the United States as of 2025 and offers **the only high containment Secure Compartmentalized Information Facility ("SCIF")**. It is a unique national asset to the federal government and is focused on two missions:

- Biological threat characterization
- Bioforensics in partnership with the Federal Bureau of Investigation (FBI) to support attribution in criminal and national security investigations

MISSION DRIVERS

NBACC's mission is to provide the scientific basis for characterization of biological threats and bioforensic analysis to support attribution of their planned or actual use.

NBACC offers unique capabilities that address security requirements outlined in HSPD-10 (Biodefense for the 21st Century), and the Homeland Security Act of 2002, Sec. 303.

NBACC OPERATIONS

NBACC is a **Government-Owned Contractor-Operated Federally Funded Research and Development Center**, currently operated by Battelle National Biodefense Institute. NBACC's daily operations include safely storing and handling biological select agents and the lab maintains compliance with the Federal Select Agents and Toxins Program and associated regulations.

To safely contain these agents, the facility includes state-of-the-art controls that require 24/7 operation, monitoring, and attention.

FOCUSED MISSION SUPPORT

NBACC has two primary mission support centers:

- The National Biological Threat Characterization Center (NTBCC) safely conducts experiments to better understand biological vulnerabilities and hazards, and to provide data that help DHS

components and interagency partners to inform their operational and strategic priorities

- The National Bioforensic Analysis Center (NBFAC) operates in partnership with the FBI's Laboratory Division and works 24/7/365 to conduct technical analyses for sensitive federal law enforcement and intelligence investigations

TECHNICAL CAPABILITIES

NBACC has several technical capabilities, including:

- Data collection on dangerous pathogens in BSL-2, 3, and 4 environments
- Characterization of viral behavior and development of informed mitigation strategies
- Analysis of sensitive forensic material for law enforcement
- Scientific Expertise in aerobiology, genomics, and bioinformatics
- Investigation of Biocrime and Bioterrorism threats
- NBACC also maintains a Work for Others (WFO) program that makes its national security capabilities available to other federal agencies

NBACC'S SERVICE TO THE NATION

Core Services – NBACC Operates as a national resource for 24/7 bioforensic support to federal law enforcement tackling the most difficult public health and biodefense challenges.

NBTCC studies the properties and behavior of biological agents, bacteria, toxins, and viruses, and uses the data collected to develop strategies for detection, prevention, and response.

NBFAC identifies, characterizes, and assesses biological components within a bioforensic sample and provides federal partners with information on any risks or threats present to aid the investigation and prosecution.

WFO provides the ability for interagency partners including Department of Defense, Department of Health and Human Services, Intelligence Communities, Department of State, United States Department of Agriculture, Department of Energy National Labs, as well as Academia, and International Partners, to leverage NBACC's biodefense capabilities.



Pictured: NBACC's laboratory facility located at Ft. Detrick, MD.

